Keshav Sharma

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EDUCATION

Vellore Institute of Technology **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) *Graduation: May 2027*

SKILL SUMMARY

Languages: Java, C/C++, Python, SQL, JavaScript, R
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks, REST APIs, Multithreading, Concurrency, Design Patterns
Backend & Frameworks: Flask, FastAPI, Node.js, React.js
Databases & Storage: MySQL, SQLite, Redis
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, VS Code, IntelliJ IDEA, Jupyter Notebook

PROJECT EXPERIENCE

***RedisLite* |LINK**

- Engineered RedisLite, a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) custom LRU cache built using a hash map and doubly linked list, with lazy and active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction using os.replace(), and automatic crash recovery; validated the system with 46 automated tests and benchmarked 619K SET ops/sec for in-memory operations and 2.8K ops/sec with durable WAL persistence.

***API Rate Limiter Library*|LINK**

- Implemented a production-ready API rate-limiting library in Python featuring five industry-standard throttling algorithms, Redis integration, and framework middleware for Flask and FastAPI applications.
- Designed a modular architecture with storage abstraction, concurrent request handling, automated testing (30 test cases), and package distribution through PyPI for public installation and reuse.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Developed a full-stack expense management platform that automates receipt processing using EasyOCR, OpenCV, and NLP-based parsing, extracting key fields such as merchant, amount, date, and category from diverse receipt formats with confidence scoring and manual correction support.
- Built and deployed a scalable application using FastAPI, React, Vite, SQLAlchemy, SQLite, JWT Authentication, and Google OAuth, featuring expense analytics, budget tracking, category-wise spending insights, receipt management (CRUD), Excel export, and a responsive dashboard hosted on Render and Vercel.

***Docwise* |LINK**

- Engineered Docwise's AI-powered PDF Q&A system using Google Gemini, enabling high-accuracy (95%+) document querying and reducing manual search time by 40%.
- Implemented TF-IDF vectorisation, chunking, and real-time processing pipelines for PDFs exceeding 200+ pages, improving system scalability and responsiveness.
- Designed and integrated chat history and session management in Streamlit, boosting usability and user retention by 35%.

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |LINK ***2026***

- Demonstrated foundational knowledge of AI, Machine Learning, Generative AI, Large Language Models, and Oracle Cloud AI services.

BCG Data Science Job Simulation |LINK ***2025***

- Performed customer churn analysis using Python (Pandas, NumPy), built and optimised a Random Forest model achieving 50% recall, and delivered data-driven business recommendations through visualisations and executive reporting.

AWS APAC Solutions Architecture virtual experience program |LINK ***2025***

- Designed a scalable AWS hosting architecture using Elastic Beanstalk and communicated deployment and cost considerations to stakeholders.

Deloitte Australia Data Analytics Job Simulation |LINK ***2025***

- Built a Tableau dashboard, performed data classification in Excel, and derived business insights through data analysis.